

Seminar in Software Engineering (MSc)

LLM based Multi-Agent-Systems for RE

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Requirements Engineering (RE) is a critical and labor-intensive process in software development. Traditionally, RE processes have been quite manual, expensive, and erroneous leading to missed requirements, inconsistency, and expensive rework. Advances in Large Language Models (LLMs) have shown promising capabilities for automating many tasks in RE. But single agent based LLMs suffer from several drawbacks including hallucinations, no domain specialization capabilities, and difficulties in coordinating complex multi-stage processes. This paper looks into using multi-agent LLMs as one possible way of overcoming these disadvantages. It evaluates how multiple specialized agents may work together to perform RE tasks better than their counterparts in traditional and single-agent approaches. In this paper, we describe the architectures of multi-agents, their application in RE processes such as requirements elicitation, requirements analysis, requirements specification, and requirements traceability. Also, we discuss some significant advantages and open issues of the technology.

Keywords: Requirements Engineering, Large Language Models, Multi-Agent Systems, Software Engineering Automation, Requirements Elicitation, Requirements Analysis

Contents

List of Tables	ii
Listings	iii
1 Introduction	1
2 Background and Fundamentals	3
2.1 Requirements Engineering	3
2.2 Large Language Models	3
2.3 Multi-Agent Systems	3
3 LLM-Based Multi-Agent Architectures for Requirements Engineering	4
3.1 Motivation for Combining LLMs with Multi-Agent Systems	4
3.2 Agent Roles and Responsibilities in RE	4
3.3 Communication Patterns and Coordination	4
3.4 Case Study: The Mare Framework	4
4 Applications in Requirements Engineering	5
4.1 Requirements Elicitation	5
4.2 Requirements Analysis and Consistency Checking	5
4.3 Requirements Specification	5
4.4 Requirements Traceability	5
4.5 Real-World Case Studies	5
4.5.1 Financial Services Onboarding System	5
4.5.2 Healthcare Information System	5
5 Advantages and Benefits	6
5.1 Efficiency Improvements	6
5.2 Quality Improvements	6
5.3 Scalability Benefits	6
5.4 Cost and Business Impact	6
6 Challenges and Limitations	7
6.1 Technical Challenges	7
6.2 Domain and Knowledge Challenges	7
6.3 Evaluation and Validation Challenges	7
6.4 Practical Deployment Challenges	7
6.5 Ethical and Trust Concerns	7
7 Conclusion	8
Bibliography	9

List of Tables

Listings

1 Introduction

Software project failures are alarmingly common in the industry. Research indicates that up to 45% of large software projects fail or significantly exceed their budgets and schedules due to inadequate requirements understanding and management [Som15, CA07]. This crisis in software development highlights the critical importance of improving how requirements are engineered and managed throughout the software development lifecycle.

Requirements Engineering (RE) is the process of discovering, analyzing, and managing software system requirements [NE00]. It forms the foundation for successful software projects because clear, well-defined requirements guide design decisions, development efforts, and testing strategies. When RE is done poorly, the consequences are severe: projects deliver features users don't need, miss critical functionality, contain inconsistencies that cause defects, and experience costly rework when issues are discovered during testing or after deployment.

Requirements traditionally have been elicited in a manual manner through extensive interviews with stakeholders, document reviews, meetings, and workshops. However, such an approach to requirement elicitation encounters several obstacles. First, requirement elicitation is a very time-intensive process and requires many hours spent on interviewing stakeholders. For medium or large-scale projects, this process can take up several weeks, if not months. Second, the requirements collected manually may suffer from ambiguities, inconsistencies, and omissions that can be hard to identify and only revealed during the later phases of development or testing. Third, keeping track of how requirements are linked with other artifacts, such as design and testing, becomes more challenging as projects become larger and more complex.

Recent advances in Artificial Intelligence, particularly Large Language Models (LLMs), have opened new possibilities for automating and improving RE tasks [Aut24b]. LLMs like GPT-4, Claude, and Gemini demonstrate remarkable capabilities in natural language understanding, information extraction, text generation, and reasoning. These capabilities could potentially be leveraged to automate requirement extraction from documents, identify inconsistencies in requirement specifications, generate formal specifications, and maintain requirement traceability which are essential part of the Requirement Engineering.

Nevertheless, there exist some significant drawbacks associated with single LLM approaches that limit the use of such models in RE. One of the key disadvantages of LLMs is that they can generate fictitious information, known as hallucinations. This aspect is crucial in RE because of its focus on accuracy. A single LLM model that attempts to solve all RE problems could face issues related to its specialization in a certain domain. Furthermore, the process of conducting RE is multifaceted and involves several operations and considerations.

Multi-agent systems, where multiple autonomous agents with different roles collaborate to solve complex problems, offer a promising alternative [Woo09]. In a multi-agent approach to RE, specialized agents can focus on specific tasks (e.g., one agent extracts requirements, another checks consistency, another formalizes specifications), collaborate through well-defined communication mechanisms, verify each other's outputs to reduce hallucination, and coordinate their efforts through a manager agent that orchestrates the overall process [Aut24a, Aut24c]. This separation of concerns and specialization could address the limitations of single-agent approaches while significantly improving RE efficiency and quality.

This paper addresses the following research questions:

1. How can LLM-based agents be organized into multi-agent systems to perform RE tasks effectively?
2. What roles do individual agents play in automating different RE tasks?
3. What are the key architectural patterns and communication mechanisms for coordinating LLM-based agents in RE?
4. What are the primary benefits of multi-agent approaches compared to single agents or traditional RE?
5. What are the main challenges and limitations that must be overcome for practical adoption?

To address these questions, this paper is organized as follows. Chapter 2 provides essential background on Requirements Engineering, Large Language Models, and Multi-Agent Systems. Chapter 3 describes the architectures and mechanisms of LLM-based multi-agent systems for RE, including agent roles, communication patterns, and case studies from recent research. Chapter 4 discusses concrete applications in various RE tasks with real-world examples. Chapter 5 analyzes the significant advantages and benefits that multi-agent systems provide. Chapter 6 critically examines the challenges and limitations that currently constrain this approach. Finally, Chapter 7 concludes with key insights and directions for future research.

2 Background and Fundamentals

This chapter provides the fundamental background on requirements engineering, large language models, and multi-agent systems.

2.1 Requirements Engineering

This section defines requirements engineering and outlines its core activities, traditional challenges, and general importance.

2.2 Large Language Models

This section introduces transformer-based large language models, their key capabilities, and their major operational limitations.

2.3 Multi-Agent Systems

This section defines autonomous agents and explains the benefits of multi-agent collaboration for complex tasks.

3 LLM-Based Multi-Agent Architectures for Requirements Engineering

This chapter examines the design, coordination, and typical architectures of multi-agent systems applied to software requirements.

3.1 Motivation for Combining LLMs with Multi-Agent Systems

This section explains how multi-agent structures mitigate individual model limitations like hallucinations and restricted context.

3.2 Agent Roles and Responsibilities in RE

This section details the roles of specialized agents like elicitation, analysis, specification, and coordinator agents.

3.3 Communication Patterns and Coordination

This section outlines how agents exchange messages, share context, and resolve conflicts during collaborative tasks.

3.4 Case Study: The Mare Framework

This section reviews the Mare framework, highlighting its multi-agent architecture and published evaluation results.

4 Applications in Requirements Engineering

This chapter explores practical use cases of multi-agent systems across key requirements engineering activities.

4.1 Requirements Elicitation

This section discusses using autonomous agents to extract requirement candidates from interview transcripts and documents.

4.2 Requirements Analysis and Consistency Checking

This section details agent-based analysis of requirements to identify conflicts, ambiguities, and missing information.

4.3 Requirements Specification

This section describes how agents automatically format and structure informal requirements into standardized system specifications.

4.4 Requirements Traceability

This section explains using specialized agents to map requirements forward to code and backward to stakeholders.

4.5 Real-World Case Studies

This section presents real-world case studies demonstrating multi-agent RE applications in industry.

4.5.1 Financial Services Onboarding System

This subsection describes applying a multi-agent framework to optimize customer onboarding software requirements.

4.5.2 Healthcare Information System

This subsection details using multi-agent systems to manage complex requirements for hospital information software.

5 Advantages and Benefits

This chapter analyzes the performance, quality, and economic advantages provided by multi-agent systems in RE.

5.1 Efficiency Improvements

This section presents metrics showing how automation accelerates requirement elicitation and reduces human labor hours.

5.2 Quality Improvements

This section highlights improvements in requirement completeness, traceability, and defect reduction using multi-agent checking.

5.3 Scalability Benefits

This section details how multi-agent architectures scale to manage hundreds of requirements across distributed teams.

5.4 Cost and Business Impact

This section summarizes the cost savings, business value, and ROI of deploying multi-agent RE tools.

6 Challenges and Limitations

6.1 Technical Challenges

6.2 Domain and Knowledge Challenges

6.3 Evaluation and Validation Challenges

6.4 Practical Deployment Challenges

6.5 Ethical and Trust Concerns

7 Conclusion

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